

Assignment – 3: Data Wrangling with pandas

Aim:Data Wrangling with pandas, Manipulate and transform data to suit the analysis

Theory:

Data wrangling, often referred to as data munging, involves the critical tasks of cleansing, reshaping, and organizing raw data into a coherent and analytical-ready structure. Given that real-world datasets frequently suffer from incompleteness, inconsistencies, or lack of structure, data wrangling is a fundamental prerequisite for any meaningful data analysis or model construction.

This assignment illustrates these concepts by generating a sample Pandas DataFrame with intentional gaps in 'Name', 'Age', and 'Score' columns. The `isnull()` function was employed to pinpoint these missing entries, guiding the subsequent cleaning actions. Specifically, missing 'Age' values were imputed using the column's mean via `fillna()`, preserving data integrity. Rows lacking 'Name' entries were systematically removed using `dropna()` to ensure dataset uniformity.

Beyond mere cleaning, data wrangling encompasses data transformation. An example of this was the creation of a 'Status' column, derived conditionally from the 'Score' values. This process of generating insightful new features from existing ones is termed column transformation, a vital aspect of feature engineering.

Furthermore, the assignment touches upon Exploratory Data Analysis (EDA), emphasizing its significance in comprehending data distributions, uncovering patterns, identifying anomalies, and facilitating informed decision-making. The discussion also briefly introduces Pandas' core data structures: Series (one-dimensional), DataFrame (two-dimensional), and the largely phased-out Panel (three-dimensional).

5. Column Transformation:

Column transformation refers to the operation of altering existing columns or deriving entirely new columns from the data already present within a dataset.

1. This involves either modifying the values inside current columns or producing novel columns.
2. It aids in feature engineering by constructing more informative variables for analysis.
3. By providing a richer representation of the underlying problem, it can enhance predictive model efficacy.

4. Examples include scaling numerical data, encoding categorical attributes, performing mathematical calculations, or applying custom functions.
5. Its utility lies in converting unrefined data into a format better suited for deep analysis or machine learning applications.

```
import pandas as pd
import numpy as np

# Create a sample DataFrame with missing values for demonstration
data = {
    'Name': ['Sarang', 'Amit', 'Rohit', 'Neha', None],
    'Age': [20, 22, np.nan, 21, 23],
    'Score': [85, np.nan, 90, 88, 75]
}
df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

print("\nIs Null Check:")
print(df.isnull().sum())

# 4d: Fill missing Age with the mean Age (Data Wrangling step)
df['Age'] = df['Age'].fillna(df['Age'].mean())

# 4c: Drop rows where Name is missing
df = df.dropna(subset=['Name'])

# 5: Column Transformation (Example: Adding a 'Status' column)
# If score > 80, Status is 'Pass', else 'Average'
df['Status'] = df['Score'].apply(lambda x: 'Pass' if x > 80 else 'Average')

print("\nWrangled DataFrame:")
print(df)
```

Original DataFrame:

	Name	Age	Score
0	Sarang	20.0	85.0
1	Amit	22.0	NaN
2	Rohit	NaN	90.0
3	Neha	21.0	88.0
4	None	23.0	75.0

Is Null Check:

Name	1
Age	1
Score	1

dtype: int64

Wrangled DataFrame:

	Name	Age	Score	Status
0	Sarang	20.0	85.0	Pass
1	Amit	22.0	NaN	Average

2	Rohit	21.5	90.0	Pass
3	Neha	21.0	88.0	Pass

1. What is data wrangling?

Data wrangling, also known as data munging, is the critical process of converting raw, often untidy data into a cleaner, more structured, and accessible format, thereby enhancing its utility for analytical purposes.

Key Aspects of Data Wrangling:

1. Encompasses the comprehensive activities of cleaning, structuring, and transforming unrefined data.
2. Aims to achieve data uniformity, utility, and proper organization.
3. Involves operations such as managing missing entries, eliminating redundant records, combining disparate datasets, and rectifying data formats.
4. Its primary objective is to make data readily available and suitable for examination.

Distinction from Data Preprocessing:

1. Data preprocessing is a subsequent, more specialized phase typically performed after wrangling.
2. Its focus is on tailoring data specifically for machine learning algorithms.
3. Techniques include data scaling, normalization, converting categorical data, and feature selection.
4. The main purpose is to adapt data into an algorithm-compatible format.

2. Importance of Exploratory Data Analysis (EDA):

Exploratory Data Analysis (EDA) is an investigative procedure where a dataset is examined and condensed to grasp its fundamental properties, frequently utilizing statistical and graphical approaches prior to model development.

Benefits of EDA:

1. Provides a concise overview of a dataset's essential attributes.
2. Leverages graphical representations, such as plots and charts.
3. Facilitates the identification of underlying trends and correlations within the data.
4. Helps in detecting unusual observations, extreme values, and inconsistencies.
5. Allows for the validation of initial hypotheses before formal analytical modeling.
6. Contributes to more informed analytical choices.
7. Commonly executed with libraries like Pandas.

3. Main Differences: Series, DataFrame, and Panel:

Series: A one-dimensional, labeled array designed to store various data types.

DataFrame: A two-dimensional, labeled tabular data structure where columns can have diverse data types, akin to a relational table or spreadsheet.

Panel: A three-dimensional data container, which is now largely superseded in contemporary Pandas by the use of MultiIndex DataFrames.

4. Explain the use of:

- a. `isnull()`: Identifies the presence of null or missing data points; it generates a boolean structure mirroring the original data's dimensions.
- b. `notnull()`: Serves as the inverse of `isnull()`; it yields `True` for existing, non-missing values.
- c. `dropna()`: Eliminates rows or columns that contain any missing data from the dataset.
- d. `fillna()`: Populates missing data entries with either a user-specified constant or an aggregated value, such as an average.

5. Column Transformation:

Column transformation is the process of modifying existing columns or creating new columns from existing data in a dataset.

1. Involves changing values within existing columns or generating new ones
2. Helps in feature engineering by creating more meaningful variables
3. Improves model performance by better representing the problem
4. Can include operations like scaling, encoding, arithmetic operations, or applying functions
5. Useful for converting raw data into a more usable form for analysis or machine learning

Conclusion:

This assignment effectively illustrates the application of data wrangling techniques to refine and reshape raw data into a coherent and interpretable structure. The exercise covered the detection and proper management of missing data points, alongside the elimination of superfluous or incomplete records. Furthermore, a new column was generated, showcasing how data transformation can enrich a dataset.

The workflow underscores the paramount importance of data preparation prior to any analytical endeavor. A clean and logically structured dataset is instrumental in yielding precise insights and boosting the predictive power of models. Consequently,

data wrangling stands as an indispensable phase in any data analysis or data science pipeline.

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